Geistlinger 2020 HGSOC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

5/5 samples (≥259 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 1 speck, 25 cells removed; iLISI median 1.00→2.23 of 5); both compartments (Embryonic 2,857 / Cycling-Metabolic 3,049) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

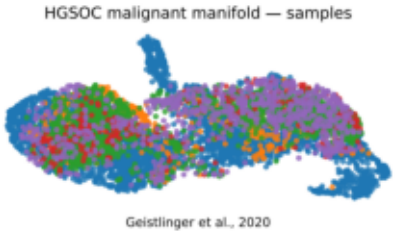
Single Gast/pSom → Crest axis: Gast & pSom co-vary (r=+0.63) and decline along pseudotime (ρ=−0.79/−0.63); Crest anti-correlated (r=−0.79/−0.95), rising to a terminal readout (ρ=+0.79; Crest-high decile mean pt=0.98). Global strength pSom>Gast>Crest-neg (Linear-pSom/Crest).

Gate 3 — cross-repression → NA

All four PPT-tree tips are pSom-leaning (no Gast arm) — the “forks” are density + cycling structure inside one co-varying programme, not a bifurcation. No Mixed hub, no branch pair to test.

STEP FIGURES

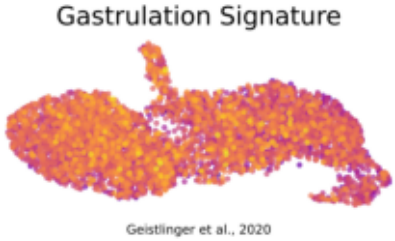
Manifold — samples



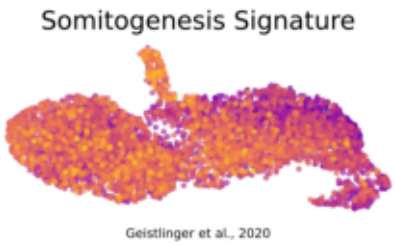
Pseudotime



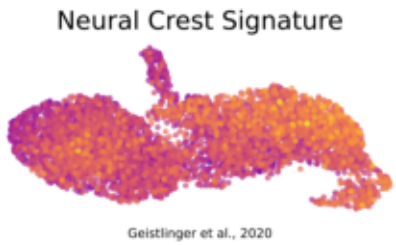
Gastrulation



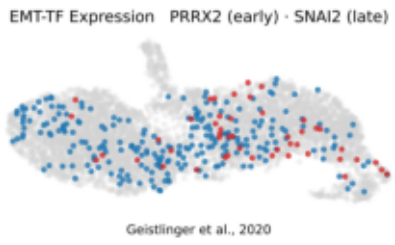
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-grade serous ovarian carcinoma (Geistlinger et al., 2020), 10x — 5 patients / 41,031 cells (master scope); malignant only: 5,931 load, 5,906 across 5 samples (≥259) after a 25-cell speck; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Geistlinger’s HGSOC malignant manifold is a single monotone Gast/pSom → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram’s hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both density lobes (cycling is a weaker secondary gradient, ρ(cyc,pt)=−0.47), so the whole manifold is kept (Bischoff/Sun/Zhou no-scoping precedent). EMT-TFs are a root/mesenchymal readout, all declining from the Gast/pSom-high root: PRRX2 (root-end, steepest) → SNAI2 (terminus-end, least declining); canonical ZEB1/TWIST near the detection floor (6.2%/2.0%).

Parameter choices

Min cells/sample	≥100 (keeps 5/5; top-of-distribution gap, no thin tail)	**Scope**	whole manifold (no scoping)
Integration	Harmony θ=10; n_pcs=20; spread=1.3	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Embryonic 2,857 / Cycling 3,049
EMT-TFs	PRRX2 (root) → SNAI2 (terminus)	**Geometry**	Linear-pSom/Crest (non-branching; fold-free 2.99)